

Automated Analyst Report

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Query: Analyse supplier performance across all product types. Identify which product categories have the highest defect rates, which shipping carriers offer the best revenue-to-cost ratio, flag any SKUs with failed inspection results, and highlight stockout risk where lead times are long relative to stock levels. report findings must be backed by solid proof from within the data

****EXECUTIVE SUMMARY**** Haircare products exhibit the highest average defect rate (2.48 %) while Carrier C delivers the strongest revenue-to-shipping-cost performance (\$1,138.56 per shipping dollar). A total of 36 SKUs failed inspection, with defect rates topping 4.94 %, and several items—most notably SKU 34 (score 13) and SKU 68 (score 8)—are at acute stockout risk due to minimal on-hand inventory versus long lead times.

****KEY FINDINGS**** - ****Product type defect hierarchy:**** Haircare = 2.483 % > Skincare = 2.335 % > Cosmetics = 1.919 % (average defect rates). - ****Carrier efficiency:**** Carrier C achieves the highest revenue per shipping dollar at ****\$1,138.56****, outperforming Carrier B (\$1,055.70) and Carrier A (\$917). - ****Inspection failures:**** ****36 SKUs**** flagged; the worst-performing SKU 42 (skincare) records a defect rate of ****4.94 %**** and only 22 units in stock. - ****Stockout exposure:**** SKU 34 (skincare) has the highest risk score of ****13**** (lead time 26 days / stock 1), followed by SKU 68 (haircare) with a score of ****8**** (lead time 8 days / stock 0). - ****Revenue concentration:**** Skincare drives the largest revenue share at ****\$241,628**** (≈ 45 % of total), while cosmetics and haircare generate \$161,521 and \$174,455 respectively.

****RISKS AND RECOMMENDATIONS**** The elevated defect rates in haircare and the concentration of failed inspection SKUs on Carrier C and high-risk SKUs (e.g., SKU 34, SKU 68) pose quality and availability threats. Immediately prioritize root cause analysis for the 36 failed SKUs, enforce stricter quality controls on haircare lines, and consider reallocating shipments from Carrier C to Carrier B where feasible to balance cost efficiency with defect mitigation. Simultaneously, increase safety stock or shorten lead times for the top-risk SKUs to avert imminent stockouts.